HI-FI AUDIO EQUIPMENT

AIWA F350

TECHNICAL GUIDE TO MECHANICS, ELECTRONICS, AND HI-FI OPERATION

> This document collects and synthesizes the detailed analysis of the physical principles, electronics, and micro-engineering mechanisms that allow the **Aiwa F350** cassette deck to deliver high-fidelity analog playback and recording.

Comprehensive Engineering & Maintenance Documentation

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Magnetic Bias and Bias Fine Adjust

Bias (or biasing current) is a high-frequency alternating current signal (typically between 80 kHz and 105 kHz, completely inaudible to the human ear) that is supplied to the recording head alongside the audio music signal.

The Physics of Bias

The magnetic particles distributed on the tape exhibit natural inertia (called *coercivity*). Without bias, weaker audio signals would fail to overcome the particles' resistance, causing severe harmonic distortion and harsh sound reproduction. Bias acts as a background vibration that keeps the magnetic particles in a state of fluid receptivity, allowing linear and faithful recording.

The "Bias Fine Adjust" Knob on the Aiwa F350

While the Aiwa F350 automatically selects the baseline bias level using mechanical sensors for cassette notches (Type I, II, and IV), the front-panel **Bias Fine Adjust** knob—equipped with a numbered scale (1–4) and a guide for various tape brands and types—allows precision tuning to compensate for chemical differences between formulations:

- **Numbered Scale (1-4) & Tape Guide:** Allows selecting the optimal recommended baseline according to the specific cassette model (e.g., TDK, Maxell, Sony, BASF).
- **Turning towards lower values (- / Less Bias):** Decreases the biasing current. The sound becomes brighter and emphasizes high frequencies (cymbals, guitars), though excessive reduction can introduce distortion.
- **Center / Reference Position (0):** Standard factory-calibrated setting for reference tapes.
- **Turning towards higher values (+ / More Bias):** Increases the current. Reduces distortion on peak signals and increases bass handling capacity, but tends to roll off high frequencies (duller sound).

Noise Reduction Systems (Dolby B and C)

The selectors labeled **B** and **C** on the Aiwa F350 belong to the noise reduction systems developed by *Dolby Laboratories*.

Dolby B

Operates primarily on mid-to-high frequencies during recording, boosting weak high-frequency signals relative to the tape background hiss. During playback, the circuit applies a mirrored attenuation, cutting background hiss by approximately **10 dB**. It is the universal standard and remains compatible with virtually all portable players.

Dolby C

Developed as an evolution of Dolby B, it uses a dual-stage compression/expansion system across two frequency bands in cascade. It delivers up to **20 dB** of noise reduction, virtually eliminating characteristic tape hiss. However, it requires the tape to be played back on decks equipped with a Dolby C decoder to avoid tonal imbalance.

Magnetic Tapes and Track Geometry

The 4 Types of Magnetic Tapes

Type	Name	Chemical Composition	Audio Characteristics
Type I	Normal / Ferro	Iron Oxide (Fe_2O_3)	Affordable, excellent for speech. Standard response with higher background hiss.
Type II	Chrome / CrO_2	Chromium Dioxide or Cobalt	Clean and bright sound, high dynamic range in highs. The '80s/'90s Hi-Fi standard.
Type III	FeCr (Ferro-Chrome)	Dual layer: Iron + Chrome	1970s experiment combining bass and highs. Rare and discontinued.
Type IV	Metal	Pure metallic iron particles	Top of the line. Extended frequency response, powerful bass, near-zero distortion.

Anatomy and Track Architecture (3.81 mm Width)

The 3.81 mm tape is physically divided into 4 parallel lanes approximately **0.6 mm wide each**, separated by safety isolation gaps (*guard bands*):

- **Side A (Upper Half):** Track 1 (Left Channel) and Track 2 (Right Channel).
- **Center Guard Band:** A 0.4–0.5 mm blank gap preventing crosstalk between sides.
- **Side B (Lower Half):** Track 3 (Right Channel) and Track 4 (Left Channel), read in the opposite tape-travel direction.

Tape Head Alignment and Azimuth

Azimuth refers to the tilt angle of the head's magnetic gap (*head gap*) relative to the tape travel axis. For optimal playback quality, this angle must be set to exactly **90° (perpendicular)**.

> **Precision Detail: Head Gap**
> The magnetic gap in the Aiwa F350 tape head measures less than **1 micron (0.001 mm)**. A misalignment of even a fraction of a degree will cause immediate loss of high frequencies (muffled sound) and phase issues between stereo channels (crosstalk).

On the Aiwa F350, azimuth is adjusted using a spring-loaded screw positioned beside the head body, enabling micrometric tilt calibration.

Slip Clutch Mechanism and Tape Transport

One of the main engineering challenges is maintaining tape travel across the head at a strictly constant speed of **4.76 cm/s**.

1. Capstan and Pinch Roller

Tape speed is governed not by the reel hubs, but by the **capstan** (a precision ground steel shaft driven by a motor and attached to a heavy flywheel) paired with a rubber **pinch roller**. Together, they clamp the tape and pull it at a precise, steady rate.

2. The Slip Clutch on the Take-Up Reel (Right Side)

As the right reel takes up tape, its diameter increases. Consequently, to gather the same length of tape (4.76 cm per second), the reel must spin progressively slower:

- At the start of a tape (small hub diameter), the reel rotates many times per second.
- Near the end of a tape (large hub diameter), the reel rotates much more slowly.

The right reel spindle is equipped with a mechanical **slip clutch**. The motor applies gentle, continuous torque, but physical resistance from the tape held by the capstan forces the clutch to "slip," allowing the reel to automatically and smoothly adjust its rotation speed in real time.

Chassis Architecture, Layout, and Muting

Bottom-Positioned Tape Heads

On stationary Hi-Fi decks like the Aiwa F350, the head assembly is located at the bottom to:

- Preserve natural cassette orientation with the label readable right-side up.
- Transfer motor and heavy flywheel weight directly to the chassis base to minimize vibration.
- Allow servomechanisms and solenoids to engage the head assembly smoothly.

Power Isolation and Low Noise Floor

Unlike portable players (Walkmans), the Aiwa F350 maintains a silent background thanks to:

1. **Isolated Power Supplies:** Distinct power lines for the motor, preamp section, and display prevent motor electrical noise from bleeding into audio circuits.
2. **Electromagnetic Shielding:** The motor is placed away from the tape heads and shielded against RF/inductive interference.
3. **Decoupled Grounds:** Dedicated ground paths prevent ground loops.

High-Inertia Flywheel and Muting Circuitry

A heavy **metal flywheel** mounted on the capstan absorbs motor micro-fluctuations through rotational inertia, minimizing *Wow & Flutter* (speed fluctuations). Finally, an electronic **Automatic Muting** circuit silences the audio path during record and pause switching, eliminating mechanical clicks and "pops" on the tape.